

The I_{3322} quantum value is attained spatially but not in finite dimension

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Abstract

Let S be the common tensor-product and commuting-operator supremum of the I_{3322} Bell functional in the Collins–Gisin normalization. Starting from a certified value window and Bellman/path equivalence, we prove that no finite-dimensional quantum strategy attains S , whereas a spatial strategy on $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ does. The finite-dimensional statement includes mixed states and binary POVMs. Consequently $C_q(3, 3; 2, 2)$ is not closed and $C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2)$ is nonempty. We also establish the dimension complexity $D(\varepsilon) = \Theta(\log(1/\varepsilon))$: approaching S requires and suffices local dimension logarithmic in inverse error. The constructive upper bound is $D(\varepsilon) \leq 23.9010650 \log(1/\varepsilon)$ for all sufficiently small ε , with natural logarithms; the lower constants are existential. The proofs use critical Bellman storage, spectral transport and normalizable orbit measures. A finite weighted-flow argument supplies the unrestricted quantitative lower bound without identifying distinct joint spectral components. This revision replaces an unsupported step in the earlier lower argument and records additional proof corrections. Prior numerical certification and independent concurrent attainment results are credited. Exact arithmetic and Lean 4 check specified numerical, scalar and finite accounting facts; the complete analytic proof is not formalized.

1 Introduction

The I_{3322} inequality [1, 2] is the best-known bipartite Bell functional beyond the Clauser–Horne–Shimony–Holt family: three binary measurements per party, and a facet of the local polytope inequivalent to CHSH. In 2010, Pál and Vértesi [3] computed its quantum value numerically to high precision, exhibited a family of finite-dimensional strategies converging to $0.2508753845\dots$, observed numerically a qubit maximum of $1/4$ and that within their construction no local dimension below 12 exceeds it, and stated in their abstract that their family “enables us to attain the largest possible quantum value in an infinite dimensional Hilbert space” [3]. They then posed the following, verbatim:

“Taking as a conjecture that our construction is optimal (in the sense that no finite dimension suffices to achieve the true quantum maximum for I_{3322}), to our knowledge this constitutes the first example of a [bipartite] Bell scenario concerning [a] finite number of measurements and outcomes, where measuring finite dimensional quantum systems is not enough to obtain exactly all quantum correlations. ... We pose it as a challenge to provide an analytical proof of our conjecture.” [3]

The parenthetical is a definition, not a second conjunct: to conjecture that the construction is “optimal” is, in their own gloss, to conjecture that no finite dimension suffices. Theorem 1 below proves exactly that statement. Their separate *assertion* of infinite-dimensional attainment is established here too, by Theorem 2, though on a route independent of their construction. Pauwels’ concurrent Theorem 1 [5] establishes equality of the full finite Pál–Vértesi-family supremum and the finite quantum supremum. This does not identify the exact value or prove convergence of a prescribed numerical parameter sequence (Section 6). Work coauthored by Vértesi in 2026 still treats the I_{3322} infinite-dimensional optimum as unresolved [30]. Coladangelo and Stark, in constructing the first explicit bipartite correlation attainable only in infinite dimension (on larger alphabets), wrote of the I_{3322} case that “an analytical proof has remained elusive” [4] (we quote the Nature Communications version).

Theorem 1 (Nonattainment). *Let $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$ be the quantum supremum in the normalization of Section 2, certified to lie in $(0.2508753845015185, 0.250875388108398]$. Then no finite-dimensional quantum strategy attains S : for all finite-dimensional Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, every state ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, and every triple of binary measurements per party — projective or POVM, acting as effects of the form $F \otimes I$ on Alice’s side and $I \otimes G$ on Bob’s — one has $\text{Tr}(\rho \mathcal{B}) < S$.*

Theorem 2 (Spatial attainment). *S is attained by a spatial strategy: there are projections $a_1, a_2, a_3 \in \mathcal{L}(\ell^2(\mathbb{Z}))$ and $b_1, b_2, b_3 \in \mathcal{L}(\ell^2(\mathbb{Z}))$, and a unit vector $\psi_S \in \ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$, such that setting $A_i = a_i \otimes I$ and $B_j = I \otimes b_j$ in (1) gives $\langle \psi_S, \mathcal{B} \psi_S \rangle = S$. In particular the resulting behavior lies in $C_{qs}(3, 3; 2, 2)$.*

Here C_q denotes the set of bipartite correlations realizable with finite-dimensional tensor-product strategies, C_{qs} its analogue allowing arbitrary separable Hilbert spaces, and $C_{qa} = \overline{C_q}$; the taxonomy is that of Paulsen, Severini, Stahlke, Todorov and Winter [31], and we follow the presentation of [21, 4].

Corollary 3. *$C_q(3, 3; 2, 2)$ is not closed, and $C_{qs}(3, 3; 2, 2) \setminus C_q(3, 3; 2, 2) \neq \emptyset$.*

The implication from I_{3322} nonattainment to nonclosure of $C_q(3, 3; 2, 2)$ is immediate from the general compactness remark of Dykema, Paulsen, and Prakash [9], who explicitly raise the I_{3322} attainment question; the corollary settles it.

Among *two-outcome bipartite* scenarios, $(3, 3; 2, 2)$ is the smallest scenario by input count in which nonclosure is known. Within that class the previously smallest was Beigi’s $(4, 4; 2, 2)$ separation of C_q from C_{qs} , which implies nonclosure of C_q there [11]; the first two-outcome nonclosure witness was the earlier $(5, 5; 2, 2)$ synchronous construction of Dykema–Paulsen–Prakash [8] (with a simplified proof by Musat–Rørdam [10]). Outside the two-outcome class the known witnesses are larger still: Beigi’s $(4, 4; 3, 3)$ nonclosure of C_{qs} [11], the $(4, 5; 3, 3)$ correlation of Coladangelo and Stark [4], the embezzlement-based nonclosure proof of [24], and the linear-system game of [13], whose input sets have sizes 184 and 235 and output sets of sizes 8 and 2.

Three inputs per party is moreover minimal among two-outcome scenarios, because a scenario in which *either* party has only two binary measurements has closed C_q . After a finite simultaneous dilation of that party’s binary effects to projections, Jordan’s lemma simultaneously block-diagonalizes that party’s two binary observables into blocks of dimension at most 2, so the behavior is a convex combination of the behaviors of the blocks. Decompose each block state into pure states; each such vector has Schmidt rank at most 2, so the other party’s

POVMs may be compressed to a two-dimensional support without changing $p(ab|xy)$. Hence $C_q(2, n; 2, 2)$ is the convex hull of a compact set, and closed, for every n ; the case $(2, 2; 2, 2)$ recovers the stronger statement that every functional there attains its maximum on two qubits [14]. Coladangelo and Stark had already singled the present scenario out: “the $(3, 3, 2, 2)$ scenario is the simplest one that is suspected to separate C_q and C_{qs} ” [4]. We do not claim minimality over scenarios with larger output alphabets.

Nonattainment opens a quantitative question that the qualitative theorems do not answer: how fast must local dimension grow as a strategy approaches S ? Section 5 settles it. Writing S_d for the optimum at local dimension at most d and $D(\varepsilon) = \min\{d : S - S_d \leq \varepsilon\}$, Theorem 6 proves $D(\varepsilon) = \Theta(\log(1/\varepsilon))$: the upper half is constructive, truncating the attaining carrier of Theorem 2 to a window of length $O(\log(1/\varepsilon))$, and the lower half shows that nothing converges faster than geometrically. The cost of approaching the unattainable value is itself a theorem.

Correction history

An earlier release of this work (3 August 2026, v1.0.0–v1.2.0 in [38]) claimed the I_{3322} principal results via a bi-infinite chain construction whose load-bearing amplitude datum was refuted by our own post-release exact audit (an exactly certified nonzero mismatch, reproduced by two independent interval engines); the claims were publicly decertified the following day (4 August, v2.0.0), and the repository’s release history preserves the full correction record. The results announced here were rebuilt on independent routes: the decertified fixed point, amplitude profile, and tail constants appear on explicit *not-used* lists in the certificate dependency records, and the failed compatibility equation is not repaired anywhere in the present chain — the new attainment route never poses it (Section 4); the reproduction recipe for every surviving claim is boxed in Section 7. Computer-assisted claims are only as credible as their failure handling, so we record this history up front.

A direct lower-bound proof in this revision. Section 5.1 replaces the previous lower-bound argument with a finite weighted-flow proof that controls all retained cell weights and accommodates spectral multiplicity without identifying state vectors in different cells. This also removes an unjustified component-identification step: the earlier paired-block identity required a pairing hypothesis not established for arbitrary retained joint spectral support. The earlier Lean cores did not formalize that operator reduction. The original files are preserved as historical artifacts, and the headline rate is unchanged in this revision; its replacement proof dates from September 2026.

Relation to independent concurrent work

The nonattainment, nonclosure, and spatial-attainment results of this paper were first announced, following the correction recorded above, in public repository releases on 5 August 2026 (git tags v3.0.0 and v3.1.0 in [38]); the dimension-complexity rate followed on 7 August 2026 (v3.3.0). Frozen archival snapshots from that week are preserved under version DOIs [10.5281/zenodo.21826916](https://doi.org/10.5281/zenodo.21826916) (v3.2.3, 6 August) and [10.5281/zenodo.21843326](https://doi.org/10.5281/zenodo.21843326) (v3.3.0, 7 August); the historical merged paper of record (v4.0.0) is archived under version DOI [10.5281/zenodo.22099128](https://doi.org/10.5281/zenodo.22099128), and the concept DOI [10.5281/zenodo.21782008](https://doi.org/10.5281/zenodo.21782008) resolves to the latest release. Independently and concurrently, Pauwels [5] proved finite-dimensional nonattainment and nonclosure by a different route (spectral-weight matrices and a Jacobi-recurrence argument), and

Coladangelo [6] independently established both finite-dimensional nonattainment and spatial attainment of the I_{3322} supremum. To our knowledge the three works were carried out independently — Pauwels states in print that nothing in his paper depends on this repository — and we regard them as concurrent. Spatial attainment is thus shared with Coladangelo’s concurrent work; the present paper gives the scalar-measure construction below and also establishes the dimension-complexity rate $D(\varepsilon) = \Theta(\log(1/\varepsilon))$. Pauwels’ Section VI records spatial attainment and the convergence-rate question as open in that work. Mghirbi’s companion paper [37] develops critical Bellman–Jacobi dynamics, spatial attainment, and logarithmic truncation of extracted carriers in a broader reversible class. Such a carrier approximation theorem and the unrestricted lower bound proved in Section 5.1 have different quantifiers; the latter must control every finite-dimensional strategy, not only truncations of a specified carrier.

How the proofs work

Both theorems run through one reduction, which we state informally here and use throughout. It is convenient to replace each party’s first two binary measurements by the pair

$$X = A_1 + A_2 - I, \quad Y = A_2 - A_1, \quad U = B_1 + B_2 - I, \quad V = B_2 - B_1,$$

which for binary projections satisfy $X^2 + Y^2 = I$ and $XY + YX = 0$, and likewise for (U, V) — the pairwise Jordan structure of a pair of binary projections. In these coordinates (1) reads

$$\mathcal{B} = X \otimes U + \frac{1}{2}X \otimes I - \frac{1}{2}I \otimes U - I + Y \otimes (B_3 - \frac{1}{2}I) + (A_3 - \frac{1}{2}I) \otimes V,$$

so the third measurements enter only through the two off-diagonal blocks. Each spectral value $t \in [-1, 1]$ of X , or of U , is a *label*. A finite word of labels $\mathbf{c} = (c_0, \dots, c_n)$ names the real symmetric tridiagonal (Jacobi) matrix $J_{\mathbf{c}}$ whose entries are explicit functions of neighboring labels — diagonal $\delta(x, u) = xu + (x - u)/2 - 1$, off-diagonal $b(t) = \sqrt{1 - t^2}/2$ — and the Pál–Vértesi block-to-Jacobi identity realizes the Rayleigh quotients of $J_{\mathbf{c}}$ as Bell values of finite strategies. Then

$$S = \sup_{\mathbf{c}} \lambda_{\max}(J_{\mathbf{c}}),$$

the same supremum for the tensor and for the commuting model: that is input (T0) of Section 2, and it is why the two values agree for this functional.

Dualizing that supremum by an LDL/Schur-pivot computation on $qI - J_{\mathbf{c}}$ replaces the word supremum by a *one-site* problem. Call a positive continuous function g on the label interval a *storage*, and call it *Bellman-feasible at level q* when

$$\frac{b(x)^2}{g(x)} + g(u) \leq q - \delta(x, u) \quad \text{for all labels } x, u.$$

Feasible storages price the steps of a path: $b(x)^2/g(x)$ is charged to the edge leaving the label x and $g(u)$ to the edge entering u . Input (T0) says that the infimum of the levels admitting a feasible storage is exactly S , and it produces one such storage g_q at each level $q > S$, namely the infimum of terminal Schur pivots over finite histories ending at the given label. The object both proofs analyze is the *critical storage* g obtained from these as $q \downarrow S$, together with the *zero set* Z on which the scalar remainder of the resulting operator decomposition vanishes — the label pairs that a strategy of value exactly S is allowed to occupy.

Theorem 1, in one paragraph. A strategy of value exactly S must occupy Z , and Z is the graph of a single strictly increasing injection P : each label is served by exactly one predecessor and serves exactly one successor. The occupied labels of a maximizer therefore carry two order-reversing maps — reflection $\sigma(u) = -u$, and $a = P^{-1} \circ (-P)$. A finite totally ordered set admits exactly one order-reversing bijection, so in finite dimension $a = \sigma$: every occupied label is paired with its own reflection. Eliminating amplitude ratios on such a reflected pair caps the value at $1/4$, and $S > 1/4$. Finite dimension enters through closure of the occupied ordered support (Remark 4).

Theorem 2, in one paragraph. The commuting model does have a maximizer, and the zero-set geometry above never used finite dimension, so the same two order-reversing maps act — now on the spectrum of the commuting pair of label operators X, U , which need not be finite. Their composition $\tau = a \circ \sigma$ is increasing and, off a null set, fixed-point-free, so each of its orbits is order-isomorphic to $\mathbb{Z}$. Disintegrating the maximizer’s scalar spectral measure over the orbits of the group generated by a and σ , and fixing one orbit, yields weights whose square roots are the amplitudes of an $\ell^2(\mathbb{Z})$ eigenvector of the Jacobi matrix at eigenvalue S . Normalizability is not imposed: it is inherited, because the weights came from a probability measure. Installing that eigenvector in the alternating block form returns a spatial strategy of value S .

Theorem 6, in one paragraph. Upper half: the labels along the carrier of Theorem 2 are monotone, so the amplitudes decay geometrically at each end, at rates κ_- and κ_+ ; truncating to a window costs only a two-bond boundary flux, and balancing the two ends gives local dimension of order $(1/\kappa_{\text{eff}}) \log(1/\varepsilon)$ with $\kappa_{\text{eff}} = (1/\kappa_- + 1/\kappa_+)^{-1}$. Lower half: the actual response operators give an approximate balance of spectral marginals on at most $4d$ labels. A Lipschitz monotone predictor separates the cyclic components of a sufficiently accurate support graph. A signed potential then controls every branch at once, forcing $\varepsilon \geq ce^{-Cd}$ without a cell-to-marginal identification.

Status of the proofs. Sections 3 and 4 retain their earlier lemma-chain presentations and source dependencies. Both quantitative arguments are given below; the new lower proof independently also implies finite-dimensional nonattainment. It does not prove spatial attainment. Section 7 distinguishes the existing Lean cores from this new, unformalized proof.

2 Setting and certified inputs

For binary effects $A_1, A_2, A_3, B_1, B_2, B_3$ (that is, $0 \leq A_i, B_j \leq I$; projections in the projective case), the Bell operator, in the normalization used throughout this paper, is

$$\mathcal{B} = -A_2 - B_1 - 2B_2 + A_1B_1 + A_1B_2 - A_1B_3 + A_2B_1 + A_2B_2 + A_2B_3 - A_3B_1 + A_3B_2, \quad (1)$$

which coincides, with no scaling or additive shift, with Eq. (3) of Pál–Vértesi [3] (equivalently, it reproduces their optimality conditions (9)–(14)). Transposing $A_1 \leftrightarrow A_2$ and $B_1 \leftrightarrow B_2$ carries (1) entry-for-entry onto the standard Collins–Gisin table — Alice marginals $(-1, 0, 0)$, Bob marginals $(-2, -1, 0)$, joint block $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ — whose local (classical deterministic) bound is 0 [2]. The two-qubit maximum is exactly $1/4$; an analytic proof for arbitrary binary qubit measurements is given in [5], Appendix A, Proposition 1. Pál–Vértesi report, as a numerical finding of their dimension scan, that no construction they found below local dimension 12 exceeds $1/4$, with their strategy family converging to $0.250875384514\dots$ [3]. We write ω_{tensor} for the supremum of $\text{Tr}(\rho \mathcal{B})$ over strategies on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\mathcal{H}_A, \mathcal{H}_B$ finite-dimensional, and $\omega_{\text{commuting}}$

for the supremum over states on the universal C^* -algebra generated by the two commuting triples. A *spatial strategy* is a tuple $(\mathcal{H}_A \otimes \mathcal{H}_B, \psi, \{a_i\}, \{b_j\})$ with $\mathcal{H}_A, \mathcal{H}_B$ separable, ψ a unit vector, and $\{a_i\}, \{b_j\}$ binary projective measurements on the respective factors acting as $a_i \otimes I$ and $I \otimes b_j$; the behaviors so realized are exactly C_{qs} . Throughout, S denotes the common value $\omega_{\text{tensor}} = \omega_{\text{commuting}}$ supplied by the certified input (T0) below, identified *only* by its certified window. We do not identify S with a historical numerical decimal or prove the limiting value of a prescribed numerical parameter sequence. The full family-supremum equality is established by Pauwels [5], Theorem 1.

Exact rational two-sided enclosures for the I_{3322} quantum value predate our own releases: Mghirbi [7] published proof-carrying exact certificates with enclosure width below 10^{-9} — tighter than the window used here — in July 2026. Those certificates establish a strict enclosure; they do not address equality of the two models, attainment, or nonattainment, which are the subject of this paper. The window below is therefore a certified *input*, not a contribution.

The proofs consume the following exactly certified inputs, established in the public repository [38]: (T0) a Bellman/path variational theorem giving $\omega_{\text{tensor}} = \omega_{\text{commuting}} = S$, together with, for every level $q > S$, a terminal-pivot storage g_q — the infimum of terminal Schur pivots over finite histories — which is positive, Bellman-feasible, and equicontinuous in q ; (LB) an exact 255-dimensional strategy certifying $S > 0.2508753845015185 > 1/4$ (rational square-root floors, independently reconstructed in 160-digit interval arithmetic); (UB) an exact rational 25,601-knot Bellman witness certifying $S \leq 0.250875388108398$; and (W) a generic operator weld: any positive Bellman-feasible storage at level q yields a positive decomposition $qI - \mathcal{B} = R_0 + R_A + R_B$. The Pál–Vértesi block-to-Jacobi identity, in its no-endpoint alternating form, supplies the bridge between Jacobi data and Bell strategies; it is a two-site local identity, established symbolically and on exact rational fixtures, and independently replicated.

These are inputs, and the dependency runs one way. In particular (T0) is certified independently of the results below, and is not re-derived from them: Theorem 2 produces a spatial strategy of value $\omega_{\text{commuting}}$, which together with $C_{qs} \subseteq \overline{C}_q$ would give $\omega_{\text{commuting}} \leq \omega_{\text{tensor}}$ — a consequence of (T0), not a premise of it.

3 Nonattainment

This section gives Theorem 1 at the level of its lemma chain: the five steps below are stated in the order they are used, each with the property of the critical storage it consumes. The complete arguments, with every constant exact, are in the proof documents of [38]; Section 7 records which parts are machine-checked.

The chain is: (N1) the critical storage exists and is concave; (N2) the two endpoints are excluded by exact rational margins; (N3) a hypothetical maximizer carries an exact equality module at S ; (N4) on that module the occupied support is a strictly increasing one-to-one graph; (N5) in finite dimension the two order-reversing transports on that graph must coincide, which caps the value at $1/4$. The critical weld below is obtained by global norm convergence, without a finite-spectrum assumption. The irreducibly finite-dimensional step is the closure argument (N5) (Remark 4).

(N1) *The critical storage.* As $q \downarrow S$ the storages g_q admit, by equicontinuity and Arzelà–Ascoli, a subsequence converging uniformly to a limit g , which is continuous, nonnegative, *concave* (each terminal pivot is affine in the target label, and an infimum of affine functions is concave), positive on the open interval, and Bellman-feasible at S . These properties, together

with the reflection-gluing inequality established below, are all that is consumed. Concavity is elementary but decisive: it forces every corner of the contact intercept downward, which is what ultimately excludes contact plateaus.

(N2) *Exact endpoint exclusion.* Two-edge finite histories give exact rational margins: taking interior detour radius $r = 1/10$ in the two-edge histories, the endpoint predecessor lines exceed the storage, uniformly over the certified window, by at least

$$m_+ = \frac{23686917837403}{3008753881083980} > 0.00787, \quad m_- = \frac{274562305945801}{4008753881083980} > 0.0684.$$

Only the positivity $m_{\pm} > 0$ is used below. Consequently the remainder R_0 has a uniform positive gap $m = \frac{1}{2} \min(m_+, m_-)$ on all endpoint spectral sectors, so any state of value at least $q - \varepsilon$ carries endpoint mass at most ε/m , an exact maximizer at S carries none, and $x = \pm 1$ is never a Bellman predecessor at S . These are closed-form rational certificates; nothing from the withdrawn construction's geometry is used.

There is also a qualitative endpoint argument that simplifies the limiting weld. One-edge histories give $g(\pm 1) \leq S < 1/3$. If $g(1) = 0$, the 3/2-Lipschitz bound and reflection gluing give $g(t) \leq \frac{3}{2}(1-t)$ and $g(-t) \geq (1+t)/6$ for $-1 < t < 1$, contradicting $g(-1) < 1/3$ as $t \uparrow 1$. Reflection treats the other endpoint, so $\min_{[-1,1]} g > 0$. This argument uses only the storage construction and gluing, not either dimension-rate bound.

(N3) *The equality module of a hypothetical maximizer.* Suppose a finite-dimensional strategy attains S . (Purification is unnecessary: $\text{Tr}(\rho R) = 0$ with $R \succeq 0$ gives $R\rho = 0$ directly. POVMs reduce to projective measurements by the extreme-effect replacement of the certificate documents: holding the state and the other effects fixed, each binary effect may be replaced by an extreme maximizer of its linear objective, and extreme binary effects are projections; this preserves the dimension and the tensor structure and does not decrease the value; since the original strategy attains S , it preserves it.) Apply the weld at levels $q_n \downarrow S$. Uniform convergence $g_{q_n} \rightarrow g$ and the strictly positive closed-interval minimum give uniform convergence of $b(t)\sqrt{g_{q_n}(-t)/g_{q_n}(t)}$ and its reflected coefficient. Functional calculus therefore gives global operator-norm convergence of all weld remainders. Since $\sum_{\nu} \text{Tr}(\rho R_{\nu,n}) = q_n - S \rightarrow 0$, the limiting positive remainders annihilate ρ . No joint spectral cutoff, or commutation of such a cutoff with the responses, is assumed. We call the common null space of those limiting remainders, which carries ρ , the *equality module*. Thus the maximizer carries an exact critical equality module at S .

(N4) *Zero-set geometry.* On the equality module, the scalar remainder symbol vanishes. Reflected finite histories give the gluing inequality $g(x)g(-x) \geq b(x)^2$ with $b(x)^2 = (1-x^2)/4$; full equality of the two Bellman inequalities and the Cauchy-Schwarz step forces $g(x)g(-x) = b(x)^2$ at every occupied pair, and a Monge (supermodularity) argument makes the full zero set monotone. The decisive step is a convex-envelope theorem: writing the contact intercept $C(x) = S + 1 - \frac{x}{2} - b(x)^2/g(x)$ on the open interval $(-1, 1)$, Bellman feasibility $g(u) \leq C(x) + (\frac{1}{2} - x)u$ evaluated at the *target* $u = 0$ gives $C(x) \geq g(0) > 0$ for every source x , so the constant $g(0)$ is a convex minorant and the greatest convex minorant $\check{C}$ exists. The strict endpoint positivity established in (N2) also makes C continuous on the closed interval. Concavity of g makes every corner of C point downward, while a kink of $\check{C}$ at a contact point would force an upward corner. Hence $\check{C}$ is differentiable, no source serves two targets, and (by an exact involution $(x, u) \mapsto (-u, -x)$ of the constraint system, which requires no symmetry of g itself) no target is served by two sources. The occupied support of any maximizer is therefore a strictly increasing one-to-one graph.

(N5) *Finiteness forces closure; closure caps the value.* The equality module carries two response isometries relating its occupied state components, with positive scalar norm ratios, implementing the decreasing involutions $a(u) = P^{-1}(-P(u))$ and $\sigma(u) = -u$ on the occupied support, where P is the increasing graph map just obtained; the graph property makes both total. A finite totally ordered set admits exactly one decreasing bijection, so $a = \sigma$ on the support: every occupied pair has its reflected partner. The resulting amplitude-ratio elimination (multiplicity-uniform: it uses the norms of the occupied components, not an operator identification of ambient joint eigenspaces) yields, for an occupied pair (x, u) ,

$$S = xu - 1 + \sqrt{(b(x) + b(u))^2 + \frac{(x-u)^2}{4}} \leq -t + \sqrt{t} \leq \frac{1}{4}, \quad t = 1 - xu,$$

contradicting $S > 1/4$. This proves Theorem 1. The nonclosure half of Corollary 3 follows by compactness of the behavior space: finite-dimensional behaviors approach value S ; a convergent subsequence has value exactly S ; closedness of C_q would give a finite-dimensional realization attaining S .

What is machine-checked here. The Lean 4 kernel checks the exact endpoint margins of (N2), the finite-closure lemma of (N5) — the single point at which finite dimensionality enters Theorem 1 — and the amplitude-elimination chain and quarter ceiling that (N5) ends with. Exact-arithmetic scripts guard the algebraic identities of (N1) and (N4), and each prints explicitly the analytic steps it does not cover: envelope existence and maximality, the gluing inequality, the limiting passage, spectral support, and operator closure (Section 7).

Remark 4. The earlier presentation used a finite interior spectrum in the limiting passage. The endpoint-positive global weld above removes that use. The closure step “a finite ordered set has one decreasing bijection” remains irreducibly finite-dimensional. Section 4 instead disintegrates an infinite occupied measure into normalizable orbits.

4 Spatial attainment

The idea. The commuting model always attains its value. The critical weld applies in its GNS representation as well. The finite closure step that capped the value at $1/4$ no longer applies, leaving room for attainment. What takes the place of a finite spectrum is a disintegration: the maximizer’s scalar spectral measure decomposes over the countable orbits of the group generated by its two response transports, and a single orbit already carries an $\ell^2(\mathbb{Z})$ eigenvector of the Jacobi matrix at eigenvalue S . The point to watch is normalizability, which here is not imposed but inherited — the amplitudes are square roots of the weights of a probability measure. The route of our withdrawn release instead constructed amplitudes along a bi-infinite chain and then demanded a global amplitude-compatibility equation, the equation our own audit refuted. Here amplitudes are extracted from conditional probability measures, which already supply summability; the former compatibility equation is not imposed.

The chain is: (S1) a maximizing commuting state exists and the zero-set geometry of Section 3 transfers to it unchanged; (S2) the response transports are realized by exact operator blocks, giving Radon–Nikodym transport laws for the maximizer’s scalar spectral measure; (S3) the response translation is fixed-point free off a null set, its orbits are order-isomorphic to $\mathbb{Z}$, and a Borel transversal is constructed explicitly; (S4) disintegration over the orbits of the group generated by the two transports yields, on one orbit, amplitudes that are automatically square-summable and satisfy the Jacobi eigenvalue equation; (S5) the Pál–Vértesi block identity installs the resulting eigenvector as a spatial strategy of value exactly S .

(S1) *A commuting maximizer, and the same zero set.* By weak-* compactness of the state space of the universal commuting C^* -algebra, a maximizing state exists for the commuting model: the commuting-operator value of a Bell functional is always a maximum [19, 20]; the nonattainment here concerns finite-dimensional tensor-product strategies. Passing to the GNS representation of that state, write Ω for the associated cyclic unit vector, so that expectations become vector expectations. The global critical weld of Section 3 applies in this representation as well, and the zero-set geometry of Section 3 transfers unchanged, since the envelope argument consumed only the storage properties and the gluing inequality, neither of which used finite dimension. The full zero set is a strictly increasing one-to-one graph P . The commuting maximizer is supported there; the transport laws in (S2) allow restriction to a common Borel conull domain invariant under all finite words in $a(u) = P^{-1}(-P(u))$ and $\sigma(u) = -u$.

(S2) *Transports without a finite spectrum.* No finite spectral support is assumed. The transports are expressed through the exact operator blocks $W = Y(B_3 - \frac{1}{2}I)$ and $W_B = (A_3 - \frac{1}{2}I)V$, which satisfy $W^2 = b(X)^2$ and $WX = -XW$, with the corresponding U -side identities for W_B , as projector identities (no global spectral involution is assumed; it need not exist). Here X and U are the commuting source and target label operators, and Y, V the corresponding reflection blocks. These yield exact Radon–Nikodym transport laws for the scalar spectral measure μ of the commuting pair (X, U) : writing E for the joint projection-valued measure of that pair, $\mu(\Delta) = \langle \Omega, E(\Delta)\Omega \rangle$ is a Borel probability measure, normalized because $\|\Omega\| = 1$. Denote its U -marginal by μ_U ; graph support transports the X -marginal law onto this same measure.

(S3) *Orbits of the response translation.* The fixed-point set of the response translation $\tau = a \circ \sigma$ carries no mass — on it the two transport densities coincide, which fires the quarter-ceiling elimination against $S > 1/4$; the set need not be empty, and nullity is all that is used. Off it τ is fixed-point-free and, being a composition of two decreasing maps, strictly increasing, so each of its orbits is order-isomorphic to $\mathbb{Z}$ and a Borel transversal can be *constructed*: select, in a fixed enumeration, the first rational strictly between the infimum and supremum of the orbit, then its unique half-open crossing. The rational selection is constant on the entire orbit. Reflection pairs one or two translation orbits, giving a Borel transversal for the group in (S4), without an appeal to a general smoothness assertion.

(S4) *Disintegration, and inherited normalizability.* Pass to the group $G = \langle a, \sigma \rangle$ generated by the two transports, which is infinite dihedral and whose orbits may carry nontrivial stabilizers: μ_U disintegrates over the countable G -orbits. The transport laws are Radon–Nikodym identities, valid off a single null set of transversal points; its complement has full measure and is therefore non-empty, and fixing one point t there, the conditional measure μ_t satisfies them at every atom of its orbit. Put $u_n = \tau^n(t)$. Interleaving the weights of this orbit of t with its reflection $\{-u_n\}$ gives, explicitly,

$$c_{2n} = u_n, \quad c_{2n+1} = -P(-u_n), \quad \tilde{\lambda}_{2n} = \sqrt{\mu_t(\{u_n\})}, \quad \tilde{\lambda}_{2n+1} = \sqrt{\mu_t(\{-u_n\})}.$$

The odd label and the reflected point supplying its amplitude must not be identified. Automatically $1 \leq \sum_j \tilde{\lambda}_j^2 \leq 2$: *normalizability is inherited from the probability measure, not imposed* — this is the precise point where the historical obstruction disappears. Set $\lambda = \tilde{\lambda}/\|\tilde{\lambda}\|_2$. The transport laws *force* the cocycle $\lambda_{j+1}/\lambda_j = g(c_j)/b(c_j)$, and the Bellman contact equality closes the Jacobi eigenvalue equation $J\lambda = S\lambda$ exactly, where J is the Jacobi matrix whose entries (diagonal δ , off-diagonal b) were given in the introduction. Normalization changes neither the cocycle nor the eigenvalue equation.

(S5) *Installing the eigenvector as a strategy.* The Pál–Vértesi block identity, in its no-

endpoint alternating form, installs the normalized eigenpair as a spatial strategy with value S ; the identity is two-site local, so it applies to the bi-infinite chain term-by-term, every Bell term being a sum over disjoint two-site blocks with diagonal sums dominated by $\sum_j \lambda_j^2$ and neighbor sums absolutely convergent by Cauchy–Schwarz. No limit is taken and no operator topology is invoked. Every finite principal block of J is itself $\preceq SI$, so the value cannot exceed S , and the eigenvalue equation attains it.

Concretely, the bipartite vector is $\psi_S = \sum_j \lambda_j e_j \otimes e_j$ in the standard basis of $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ — a unit vector by the rescaling in (S4), as Theorem 2 requires; the alternating block form installs a_1, a_2, a_3 on the left factor and b_1, b_2, b_3 on the right, each an orthogonal direct sum of rank-one projections over one of the two perfect matchings of $\mathbb{Z}$, so that $A_i = a_i \otimes I$ and $B_j = I \otimes b_j$ as Theorem 2 requires. The block *form* here is Pál–Vértési’s; the labels c_j and the amplitudes λ_j come from the disintegration. This proves Theorem 2 and, with Theorem 1, the separation in Corollary 3.

What is machine-checked here. None of it. The measure-theoretic and operator-algebraic chains of this section are not formalized in Lean 4, and the exact-arithmetic guard scripts name the limiting passage, spectral support and operator closure explicitly among the steps they do not cover. They are proved here and in the certificate’s proof document, and were audited in the review rounds recorded there (Section 7).

The measure-theoretic steps of this argument — the conull invariant set, the Borel transversal, and the uniqueness of the disintegration (§§6–9 of the certificate’s proof document) — were originally flagged in the certificate’s status file as carrying residual proof risk. An expanded argument was supplied on 7 August 2026 in a full write-up (25 numbered lemmas, each with complete proof and explicit quantifier labels, plus an axiom inventory) reviewed in two blind adversarial rounds: a hostile proof-surface review together with an independent countermodel search (22 constructed attacks, no counterexample), followed by a re-review of the repaired document that verified every repair item by item (AMENDMENT-2026-08-07-SECTIONS-6-9 in the certificate directory). The status file itself is hash-frozen and deliberately unamended; the amendment is the correction of record.

Remark 5 (Envelope versus orbit). The closure argument of Section 3 forces the $u \mapsto -u$ symmetry only on the *occupied orbit* of a hypothetical finite maximizer; nothing forces the critical envelope itself to be reflection-symmetric, and no step above obtains one wing of the envelope by reflecting the other. The historical construction’s implicit reflection symmetry is, in retrospect, adjacent to the exact incompatibility that forced its withdrawal.

5 Dimension complexity

This section determines how fast the finite-dimensional optima approach S . Throughout, $\log$ denotes the natural logarithm; and, in this section only, $\mu(t)$ denotes the band multiplier (not the spectral measure of Section 4), $\Delta(t)$ the diagonal cost $\delta(t, t)$ (not a Borel set). The symbol I keeps its meaning as the identity operator throughout, and windows of integers are written $\mathcal{I}$. Define, for each $d \geq 1$,

$$S_d := \sup \left\{ \langle \psi, \mathcal{B} \psi \rangle : \dim \mathcal{H}_A \leq d, \dim \mathcal{H}_B \leq d, \text{ three projection-valued binary measurements per party on } \mathcal{H}_A, \mathcal{H}_B, \psi \in \mathcal{H}_A \otimes \mathcal{H}_B \text{ a unit vector} \right\},$$

$$D(\varepsilon) := \min \{ d \in \mathbb{N} : S - S_d \leq \varepsilon \}.$$

The parameter set is compact — a finite product of projection varieties on $\mathbb{C}^d$ together with the unit sphere of $\mathbb{C}^d \otimes \mathbb{C}^d$ — and the value is continuous, so the supremum defining S_d is attained; Theorem 1 therefore gives $S_d < S$ for every d . S_d is nondecreasing, and $S_d \rightarrow S$ by the construction below, so $D(\varepsilon)$ is well-defined and finite for every $\varepsilon > 0$. (The cap is symmetric on the two parties; the construction below delivers equal local dimensions, so the same function results.) The class is projective measurements and pure states, with the dimension cap imposed on the local Hilbert spaces and no dilation; this loses no generality for the value: the objective is affine in the state, so an extreme — pure — state attains it, and affine in each binary effect separately, so each may be replaced by an extreme effect, which is a projection — the same replacement as in Section 3, and a specifically two-outcome fact. Naimark dilation is never invoked, so the local dimensions are unchanged [33]; in particular the POVM-indexed rate is the same function.

Possible asymptotic scales include $D(\varepsilon) = \Theta(\log(1/\varepsilon))$ — geometric convergence; or $D(\varepsilon)/\log(1/\varepsilon) \rightarrow \infty$, of which polynomial dimension cost $D(\varepsilon) \sim \varepsilon^{-c}$ is one instance and the $2^{\Omega(\varepsilon^{-1/8})}$ of [24] a far more expensive one; or $D(\varepsilon) = o(\log(1/\varepsilon))$, which super-geometric convergence such as $S - S_d \sim e^{-d^2}$ would produce. The following theorem proves the logarithmic scale, and it takes both halves to do so: windowing the attaining carrier yields strategies that converge at least geometrically, and — by the lower bound — nothing converges faster.

Theorem 6 (Rate). *There are constants $c_1, c_2 > 0$ with $c_1 \log(1/\varepsilon) \leq D(\varepsilon) \leq c_2 \log(1/\varepsilon)$ for all sufficiently small $\varepsilon > 0$. The upper half holds with the explicit derived constant: $D(\varepsilon) \leq 23.9010650 \log(1/\varepsilon)$ for all sufficiently small ε .*

Where the two halves are proved. The lower proof is in §5.1, and the constructive upper proof is in §5.2. Both use the fixed critical storage and compact interiority derived below. Only the upper proof consumes the spatial carrier and its strict graph structure.

In gap form, given the monotonicity of S_d : there is $\kappa_{\text{eff}} > 0$ (defined in §5.2, with $1/\kappa_{\text{eff}} \leq 23.9010650$) such that for every $0 < \alpha < \kappa_{\text{eff}}$ there is d_0 with $S - S_d \leq e^{-\alpha d}$ for all $d \geq d_0$; and there are $c, C > 0$ with $S - S_d \geq ce^{-Cd}$ for every d .

5.1 The lower bound by finite weighted flow

We prove, with constants depending only on the fixed Bell functional,

$$S - S_d \geq c_* e^{-C_* d} \quad (d \geq 1). \quad (2)$$

The proof uses neither a dimension-preserving conversion to a chain nor identification of vectors in different joint spectral cells. There can be quadratically many such cells. We count only the distinct local spectral labels, and control all nonnegative cell weights simultaneously.

Critical data and the original-dimension interface

Write

$$\delta(x, u) = xu + (x - u)/2 - 1, \quad b(t) = \frac{1}{2}\sqrt{1 - t^2}, \quad w = b^2.$$

The critical-storage construction of Section 2 gives a concave $3/2$ -Lipschitz function g on $[-1, 1]$, positive in the open interval, such that

$$\frac{w(x)}{g(x)} + g(u) \leq S - \delta(x, u), \quad g(t)g(-t) \geq w(t). \quad (3)$$

Initially the quotient is taken at interior points. To make the inherited input precise, for $q > S$ take the infimum g_q of terminal Schur pivots over finite Jacobi histories. The bound $J \preceq SI$ gives pivots at least $q - S$; extending a history gives the first inequality at level q . Each terminal-pivot function has slope $1/2 - x$ in the terminal label, so their infimum is concave and $3/2$ -Lipschitz. Monotonicity in q and the common modulus give a uniform limit g as $q \downarrow S$. Feasibility with a fixed target label bounds $w(t)/g_q(t)$ uniformly for each interior t , proving $g(t) > 0$ there. Concatenating a history ending at t with the reflected reverse of a history ending at $-t$ gives two central Schur pivots coupled by $b(t)$. The resulting positive 2×2 matrix gives the second inequality in (3) after taking the two infima and the limit. This is the critical-storage input, not a consequence of the old finite-dimensional rate argument. For a related treatment of canonical storage and positive welds for reversible Bell functionals, see Mghirbi [37]. The argument below supplies the unrestricted lower bound directly; no such bound is imported from that work.

Here is an elementary endpoint check using only $S < 1/3$. One-edge histories give $g(1), g(-1) \leq S$. If $g(1) = 0$, Lipschitz continuity and reflection gluing imply, for $-1 < t < 1$,

$$g(t) \leq \frac{3}{2}(1-t), \quad g(-t) \geq \frac{1-t^2}{4g(t)} \geq \frac{1+t}{6}.$$

Letting $t \uparrow 1$ contradicts $g(-1) \leq S < 1/3$. Reflecting proves $g(-1) > 0$ also. Thus $m := \min_{[-1,1]} g > 0$, and all quotients below extend continuously to the endpoints.

Set

$$p = w/g, \quad \alpha(t) = b(t)\sqrt{g(-t)/g(t)}, \quad \beta(t) = \alpha(-t), \quad \phi(x, u) = S - \delta(x, u) - \alpha(x) - \beta(u).$$

Feasibility and its reflected version, followed by Cauchy–Schwarz, give $\phi \geq 0$. In detail, the two vectors $(\sqrt{p(x)}, \sqrt{g(u)})$ and $(\sqrt{g(-x)}, \sqrt{p(-u)})$ have inner product $\alpha(x) + \beta(u)$ and squared norms at most $S - \delta(x, u)$. At $x = \pm 1$ they cannot be proportional: the first coordinate of the first vector vanishes, whereas both $g(u)$ and $g(-x)$ are positive. Consequently $\phi > 0$ on those two boundary edges; reflection treats the other two. Compactness supplies a symmetric interval $K = [-a, a] \subset (-1, 1)$ containing every zero strictly inside K^2 , and a constant $a_0 > 0$ with

$$\inf_{(x,u) \notin K^2} \phi(x, u) \geq a_0. \quad (4)$$

For example, minimize on the closed boundary collar $\max(|x|, |u|) \geq a$. No minimum on a nonclosed set is needed.

First reduce arbitrary binary effects and mixed states without dimension loss. At fixed state the Bell value is affine in each effect; successively maximize each coordinate at a projection, then choose a top eigenvector of the resulting Bell operator. The value cannot decrease and neither local dimension changes. It therefore suffices to prove the lower bound for projective pure-state strategies. For such measurements on the original tensor product, put

$$X = A_1 + A_2 - I, \quad Y = A_2 - A_1, \quad U = B_1 + B_2 - I, \quad V = B_2 - B_1,$$

and $W_A = Y(B_3 - I/2)$, $W_B = (A_3 - I/2)V$. Direct multiplication, using idempotence of the measurement projections, gives

$$W_A^* = W_A, \quad W_A X = -X W_A, \quad W_A^2 = b(X)^2,$$

and the analogous identities for W_B, U . In particular W_A is not assumed to commute with U . The operators $Q_A = \alpha(X) - W_A$ and $T_A = \alpha(-X) + W_A$ are self-adjoint and satisfy $Q_A T_A = 0$:

use $W_A f(X) = f(-X)W_A$ and $\alpha(t)\alpha(-t) = b(t)^2$. They therefore commute. Their sum is nonnegative, so their joint spectral values show that both factors are nonnegative. Likewise $Q_B = \beta(U) - W_B \geq 0$. Expanding gives

$$SI - \mathcal{B} = \phi(X, U) + Q_A + Q_B. \quad (5)$$

For a unit vector of deficit $\varepsilon = S - \langle \mathcal{B} \rangle$ and its joint spectral probability measure μ , positivity and bounded norms imply

$$\int \phi d\mu \leq \varepsilon, \quad \|Q_A \psi\| + \|Q_B \psi\| \leq C\sqrt{\varepsilon}, \quad \mu((K^2)^c) \leq a_0^{-1}\varepsilon. \quad (6)$$

Indeed $\|Q_A \psi\|^2 \leq \|Q_A\| \langle Q_A \rangle$, with a fixed, dimension-independent norm bound. This derives the approximate response estimate from the positive weld, not from an exact-equality statement.

If the original local dimensions are at most d , each of X, U has at most d distinct eigenvalues, regardless of spectral multiplicities.

A quantitative monotone predictor

Put $C_0(x) = S + 1 - x/2 - p(x)$ and let H be its greatest convex minorant on the full interval $[-1, 1]$. There is a fixed $M \geq 1$ such that $h = H'$ is nondecreasing and M -Lipschitz in the interior, with a continuous constant extension past either endpoint. For completeness, for smooth positive concave approximants to g ,

$$p'' = \frac{w''}{g} - \frac{2w'g'}{g^2} + \frac{2w(g')^2}{g^3} - \frac{wg''}{g^2} \geq -\frac{1}{m} - \frac{4L}{m^2},$$

where L bounds the slopes and the approximants are at least $m/2$. Concave affine extensions followed by mollification and uniform limits show that C_0 is semiconcave. One may take $M = 1 + 1/m + 4L/m^2$. On every open gap where $H < C_0$, maximality makes H affine. At an interior contact,

$$C'_{0,-} \leq H'_- \leq H'_+ \leq C'_{0,+},$$

whereas semiconcavity reverses the outside inequality. All four values coincide. Adding the semiconcavity inequalities at contact points $x < y$ gives $0 \leq H'(y) - H'(x) \leq M(y - x)$. The affine gaps extend this estimate to arbitrary points, establishing the assertion.

The affine minorant at $\min_z(C_0(z) - uz)$ lies below H , so

$$g(u) - u/2 \leq \min_z(C_0(z) - uz) = \min_z(H(z) - uz).$$

Write $D = S - \delta(x, u)$, $L_1 = D - p(x) - g(u)$ and $L_2 = D - g(-x) - p(-u)$. Both are nonnegative, and

$$\alpha(x) + \beta(u) \leq \sqrt{(D - L_1)(D - L_2)} \leq D - (L_1 + L_2)/2.$$

Consequently

$$H(x) - ux - \min_z(H(z) - uz) \leq L_1 \leq 2\phi(x, u). \quad (7)$$

Choose $\rho > 0$ with every $x \in K$ at least ρ from the interval endpoints. For $F(z) = H(z) - uz$, the admissible step $z = x - \rho \operatorname{sign} F'(x)$ shows that a gap smaller than $M\rho^2/2$ implies $|F'(x)| < M\rho$. The full step $x - F'(x)/M$ is then admissible; the smoothness inequality gives a decrease at least $|F'(x)|^2/(2M)$. Thus, for every s below a fixed sufficiently small positive threshold,

$$x, u \in K, \phi(x, u) \leq s \implies |u - h(x)| \leq 2\sqrt{Ms}. \quad (8)$$

Scalar balance without identification of joint cells

Let p_X, p_U be the original spectral marginals, and put $r(t) = \sqrt{g(-t)/g(t)}$, $R(t) = r(t)^2$. On K , R is Lipschitz, with $0 < R_{\min} \leq R \leq R_{\max} < \infty$. The operator $b(X)^{-1}W_A$ is an isometry on the X -corridor and reflects its spectral projections. Projecting the response residual in (6), dividing by $b(t) \geq b_0 > 0$, and using the reverse triangle inequality yields

$$\sum_{t \in K} \left(\sqrt{p_X(-t)} - r(t) \sqrt{p_X(t)} \right)^2 \leq C\varepsilon.$$

Missing spectral atoms have weight zero. Cauchy–Schwarz converts this to the total ℓ^1 bounds

$$\|\mathcal{F}p_X - Rp_X\|_{1,K} \leq C_1\sqrt{\varepsilon}, \quad \|\mathcal{F}p_U - R^{-1}p_U\|_{1,K} \leq C_2\sqrt{\varepsilon},$$

where $\mathcal{F}$ reflects a measure. These inequalities use genuine projected response vectors but not their localization in joint cells.

Average measures only: $\bar{\mu} = (\mu + J_*\mu)/2$, $J(x, u) = (-u, -x)$. Since $\phi \circ J = \phi$, the defect bound is unchanged. Its marginals $p = (p_X + \mathcal{F}p_U)/2$, $q = (p_U + \mathcal{F}p_X)/2$ satisfy

$$q - Rp = \frac{1}{2}(\mathcal{F}p_X - Rp_X) - \frac{1}{2}R(\mathcal{F}p_U - R^{-1}p_U).$$

Fix the vertex universe to be the union of the original and reflected spectral labels in K , of cardinality $n \leq 4d$. Retain only cells $(x, u) \in K^2$ with $\phi(x, u) \leq s \leq 1$, as directed edges $x \rightarrow u$. Keep isolated vertices. If m_s is retained mass and o, i are outgoing and incoming masses, Markov's inequality gives fixed constants with

$$1 - m_s \leq D_0\varepsilon/s, \quad \|i - Ro\|_1 \leq C_b\sqrt{\varepsilon} + (1 + R_{\max})D_0\varepsilon/s. \quad (9)$$

No physical realization of $\bar{\mu}$ is asserted or needed.

Cycles and paths

An elementary monotonicity observation will be used. If $x_0, \dots, x_k = x_0$ satisfies $|x_{j+1} - h(x_j)| \leq \delta$ with h nondecreasing, then $|h(t) - t| \leq \delta$ throughout the visited interval. Every interior level is crossed upwards and downwards; an upward crossing gives $h(t) \geq t - \delta$, and a downward one gives $h(t) \leq t + \delta$. Predecessor and successor inequalities give the same bounds at the minimum and maximum. In particular every cycle edge has length at most 2δ .

Put $\Delta = S - 1/4 > 0$. Since $t^2 - 1 + 2b(t) \leq 1/4$, on the diagonal

$$\phi(t, t) = S - [t^2 - 1 + b(t)(r(t) + r(t)^{-1})].$$

For $|R(t) - 1| \leq c \leq 1/2$ the identity $r + r^{-1} - 2 = (R - 1)^2/[r(r + 1)^2]$ implies $\phi(t, t) \geq \Delta - c^2$. Choose $c = \min(1/2, \sqrt{\Delta}/2)$. On a retained cycle, (8) instead bounds $\phi(x, x) \leq s + 2L_\phi\delta$, where $\delta = 2\sqrt{M}s$ and L_ϕ is a Lipschitz constant on K^2 . Choose a fixed sufficiently small $\delta_* > 0$ so that the latter quantity is at most $\Delta/2$ whenever $\delta \leq \delta_*$. Every cyclic vertex must then have $R \leq 1 - c$ or $R \geq 1 + c$. Every internal edge of a cyclic strongly connected component belongs to a cycle. Decreasing δ_* so that $2L_R\delta_* < 2c$ shows that such a component is entirely low or entirely high.

Individual components are not enough: a low cycle feeding a high cycle can support a nonzero exactly balanced flow. To exclude this, also take $\delta_* \leq c/(4 \max(1, L_R))$, and choose the cut using the previously fixed universe:

$$\delta_n = \delta_*(M+1)^{-n}, \quad s_n = \delta_n^2/(4M). \quad (10)$$

Starting at a cyclic vertex a , any k -edge path satisfies

$$|x_{j+1} - a| \leq M|x_j - a| + 2\delta_n, \quad |x_k - a| \leq 2\delta_n \sum_{j=0}^{k-1} M^j \leq 2\delta_n(M+1)^k.$$

A shortest path is simple, hence has at most $n-1$ edges. Its endpoint cannot be an opposite-gain cyclic vertex: such vertices differ in R by at least $2c$, while the displayed distance and Lipschitz continuity give a difference at most $c/2$. In particular no low cyclic component can reach a high cyclic component.

A signed potential that controls every branch

For any n -vertex directed graph with these properties there is a real function f such that

$$R_x f_x - f_y \geq 1 \quad (x \rightarrow y), \quad \|f\|_\infty \leq B_*^{n+1}, \quad B_* = 4 + 2R_{\max} + 2/R_{\min} + 2/c. \quad (11)$$

Here is the construction. Let $\mathcal{U}$ be all vertices reachable from low cyclic components. It is forward closed and contains no high cycle. Its complement is backward closed, and no strongly connected component straddles the split: any component meeting $\mathcal{U}$ is contained in it by forward closure. On its condensation directed acyclic graph choose constants in topological order,

$$g_C = \max\{1, 1/c, \max_{D \rightarrow C} (R_{\max} g_D + 1)\},$$

omitting empty inner maxima, and put $f = -g_C$ on C . An internal edge in a cyclic component has $(1 - R_x)g_C \geq 1$; edges between components satisfy $g_{C(y)} - R_x g_{C(x)} \geq 1$ by construction. An acyclic singleton has no internal edge. On the complementary condensation graph process in reverse topological order and set

$$f_C = \max\{1/R_{\min}, 1/c, \max_{C \rightarrow D, D \not\subset \mathcal{U}} (f_D + 1)/R_{\min}\}.$$

Its cyclic components are high, so internal edges obey $(R_x - 1)f_C \geq 1$; edges between components obey the same required inequality by the recursion. Edges into $\mathcal{U}$ obey it because $f_x \geq 1/R_{\min}$ and $f_y < 0$. No edge leaves $\mathcal{U}$. At most n stages, with fixed bounded growth at each, give the norm bound in (11).

For arbitrary nonnegative weights on all retained edges, summing gives

$$m_s \leq \sum_{x \rightarrow y} \bar{\mu}_{xy} (R_x f_x - f_y) = \sum_x f_x (R_x o_x - i_x) \leq B_*^{n+1} \|Ro - i\|_1. \quad (12)$$

This is why branching and spectral multiplicity introduce no missing payment: no edge, row, or component is replaced by a selected path.

Assembly

If the fixed universe is empty, the collar estimate already gives a positive lower bound on ε . Otherwise combine (9)–(12), with $C_3 = (1 + R_{\max})D_0$ and constants enlarged to at least one:

$$1 - D_0\varepsilon/s_n \leq B_*^{n+1}(C_b\sqrt{\varepsilon} + C_3\varepsilon/s_n).$$

Either $D_0\varepsilon/s_n \geq 1/2$, or one of the two right-hand terms is at least $1/4$. Hence

$$\varepsilon \geq \min \left\{ \frac{s_n}{2D_0}, \frac{1}{16C_b^2B_*^{2n+2}}, \frac{s_n}{4C_3B_*^{n+1}} \right\}.$$

By (10) this is at least $c_0e^{-C_0n}$ with fixed positive constants. Since $n \leq 4d$, (2) follows. It also excludes exact finite-dimensional attainment without using spatial attainment or the old common-return argument. Finally, $\varepsilon \geq c_*e^{-C_*d}$ implies $d \geq C_*^{-1}\log(c_*/\varepsilon)$, proving the lower half of Theorem 6. $\square$

Verification scope. This proof is analytic. Exact finite-graph tests exercise the signed potential and negative controls, but do not prove the analytic estimates. The Lean module `I3322Kernel.WeightedFlow` verifies the finite all-edge summation identity, the residual bound conditional on an edge potential, three local potential inequalities, and the elementary assembly alternative. It does not prove the existence of the potential from strongly connected components, convex-envelope regularity, or the quantum-to-flow reduction. The full chain remains an analytic proof. The earlier staircase and selected-walk cores are not dependencies of this replacement.

5.2 The upper bound, explicitly

The upper half is constructive, and is given here in compressed form: the construction, the constants, and the route are all stated, while the full accounting of the endpoint-diagonal payment and its constants is carried by the proof document [35]. It consumes Theorem 2 and the inputs named below; the reductions, the exact statements of the transported inputs, and the assembly are not formalized (§5.3).

The carrier. The proof of Theorem 2 supplies a positive unit vector $\lambda \in \ell^2(\mathbb{Z})$ — the *carrier* — and a bi-infinite label sequence $(c_j) \subset (-1, 1)$ with:

- (i) the certificate’s interior zero locus Z (the zero set of its scalar weld functional) is the graph of a strictly increasing one-to-one Borel map P , and $P(c_{j+1}) = c_j$ for every j — one map, every link (Theorem 2 certificate, §§6 and 10);
- (ii) the transport law $\lambda_{j+1}/\lambda_j = g(c_j)/b(c_j)$ (Theorem 2 certificate, §11), where $b(t) = \sqrt{1-t^2}/2$ and g is the critical storage of Section 3, continuous and positive on the open interval (a uniform positive bound on the compact label set follows below);
- (iii) the eigen-equation $J\lambda = S\lambda$ for the Jacobi matrix $J_{jj} = \delta(c_{j-1}, c_j)$, $J_{j-1,j} = b(c_{j-1})$, where $\delta(x, u) := xu + (x - u)/2 - 1$ is the certificate’s diagonal cost;
- (iv) every adjacent label pair lies in Z , and Z is compactly interior, $Z \subset\subset (-1, 1)^2$.

Item (i) is proved in the certificate's §§6–9 (conull invariant set, Borel transversal, uniqueness of disintegration) and §10; an expanded write-up of §§6–9 is included in the same certificate directory as `AMENDMENT-2026-08-07-SECTIONS-6-9`. The compact interiority (iv) is the endpoint-positivity statement of [38], used here exactly as established there and with the status recorded in §5.3. The positivity in (ii), by contrast, needs no such input on the closed interval: what the argument below uses is a positive lower bound on the compact label set only, and that follows from Section 3's own properties of g — continuous, positive on the open interval — once (iv) confines the labels to a compact subset of it. Let K be the union of the two coordinate projections of the closure of Z ; by (iv) it is a compact subset of $(-1, 1)$; all labels and label limits lie in K , and on K the functions b , g , $r := g/b$ are continuous with $b \geq b_0 > 0$, $g \geq m_g > 0$, and r , $1/r$ bounded by $R_{\max} < \infty$.

Monotone labels and geometric tails. Since P is strictly increasing and one-to-one with $P(c_{j+1}) = c_j$ for every j , the labels satisfy $c_{j+1} = P^{-1}(c_j)$ and $c_{j-1} = P(c_j)$: the orbit is functional in both directions, so it is strictly increasing, strictly decreasing, or constant, and constancy at a single adjacent pair propagates to the whole sequence. The constant case is excluded: a constant label c would make λ exactly geometric with ratio $r(c)$ — not in $\ell^2(\mathbb{Z})$ in one direction if $r(c) \neq 1$, and a non-normalizable constant vector if $r(c) = 1$. Hence the whole sequence is strictly monotone, confined to the compact K , so both tail limits $t_{\pm} \in K$ exist, and by the transport law and continuity of r the outward one-step ratios converge:

$$y_+ := \lim_{j \rightarrow +\infty} \lambda_{j+1}/\lambda_j = r(t_+), \quad y_- := \lim_{j \rightarrow -\infty} \lambda_{j-1}/\lambda_j = 1/r(t_-).$$

Dividing row j of $J\lambda = S\lambda$ by λ_j and passing to either limit gives

$$y + 1/y = \mu(t) := \frac{S - \Delta(t)}{b(t)}, \quad \Delta(t) := \delta(t, t) = t^2 - 1, \quad (13)$$

at $t = t_{\pm}$. The elementary band identity is, for $s = \sqrt{1 - t^2}$,

$$\Delta(t) + 2b(t) = t^2 - 1 + s = s(1 - s) \leq 1/4 \quad (14)$$

(machine-checked: `band_identity`, `s_mul_one_sub_s_le_quarter`, `band_quarter_ceiling`). Together with $b \leq 1/2$ (`amplitude_b_le_half`) and the certified lower window endpoint $S_{\text{LO}} := 0.2508753845015185 < S$ it yields, for every interior t ,

$$\begin{aligned} \mu(t) - 2 &= (S - \Delta - 2b)/b \geq (S - 1/4)/b \geq 2(S - 1/4) \\ &> 2(S_{\text{LO}} - 1/4) = 0.001750769003037, \\ \text{so } \mu(t) &\geq \mu_{\min} := 2.001750769003037. \end{aligned} \quad (15)$$

Since ℓ^2 excludes the growing root of (13) at each end, both outward ratios equal the decaying root $x_{\text{dec}}(\mu) = 2/(\mu + \sqrt{\mu^2 - 4})$, strictly decreasing in μ , so uniformly

$$y_{\pm} \leq x_{\text{dec}}(\mu_{\min}) \leq 0.9590241, \quad y_{\pm}^2 \leq 0.91972725 < 1. \quad (16)$$

In particular $r(t_+) = y_+ < 1$ while $r(t_-) = 1/y_- > 1$: the transport ratio crosses 1 strictly between the two tail limits — a nontrivial corroboration of the two-ended open-chain picture. Tail mass sums amplitude squares, so with $\kappa_{\pm} := -2 \log y_{\pm}$ the tails obey $\lambda_j^2 = \exp(-\kappa_{\pm}|j| + o(|j|))$ at the respective ends. In particular, for every $0 < \eta < \min(\kappa_-, \kappa_+)$ there is $A_{\eta} < \infty$

with $\lambda_j^2 \leq A_\eta e^{-(\kappa_\pm - \eta)|j|}$ on the corresponding tail. Ratio convergence alone does not justify a fixed prefactor at the limiting exponent. Carrying the down-rounded surrogate $K_0 := 0.08367827985 \leq -2 \log x_{\text{dec}}(\mu_{\min})$ (computed against the exact $x_{\text{dec}}(\mu_{\min})$, not the rounded display in (16), which is too weak to support it), which is safe both in $K_0/2$ (rounded down) and in the denominator of $2/K_0$ (a down-rounded denominator over-estimates the quotient, the safe direction for an upper display):

$$\begin{aligned} \kappa_\pm &\geq K_0 = 0.08367827985 \text{ per index,} \\ \kappa_{\text{eff}} &:= (1/\kappa_- + 1/\kappa_+)^{-1} \geq K_0/2 \geq 0.0418391, \\ 1/\kappa_{\text{eff}} &\leq 2/K_0 = 23.90106493088 \dots \leq 23.9010650. \end{aligned} \tag{17}$$

(All displayed decimals are bounds rounded in the safe direction and re-verified in exact rational arithmetic by the repository's guard scripts, which re-derive every displayed constant from its stated inputs and fail on any mismatch. Because $1/\kappa_{\text{eff}} \leq 2/K_0 = 23.90106493088 \dots$ sits strictly below 23.9010650 with a genuine gap, the eventual bound $D(\varepsilon) \leq 23.9010650 \cdot \log(1/\varepsilon)$ follows — a bare $\limsup \leq 23.9010650$ alone would not give it.)

The truncation, at exact local dimension d . For a window $\mathcal{I} = [-L, R] \cap \mathbb{Z}$, keep every full 2×2 rank-one block of each of the carrier's two alternating matchings whose paired indices lie in $\mathcal{I}$; at a severed endpoint pair, replace the block by the one-dimensional projector on the retained index (either of the two admissible one-dimensional assignments serves; we fix this choice, and the choice is part of the endpoint-diagonal payment accounted below). All six measurement operators remain exact projections — each is a direct sum of orthogonal rank-one blocks and one-dimensional projectors (verified symbolically at arbitrary block angles for window sizes $m = 3, \dots, 8$ by an independently written script; the general case is the direct-sum structure itself) — and the local dimension is exactly $d = |\mathcal{I}|$, with no dilation and no padding. The truncated state is the renormalized window of λ .

Error accounting. Let $v_{\mathcal{I}}$ be the value of the principal Jacobi compression on $\mathcal{I}$ and $V_{\mathcal{I}}$ the Bell value of the truncated strategy just built. The exact two-boundary flux identity (two lines from $J\lambda = S\lambda$):

$$S - v_{\mathcal{I}} = B_{\mathcal{I}}/M_{\mathcal{I}}, \quad B_{\mathcal{I}} := b(c_{-L-1})\lambda_{-L-1}\lambda_{-L} + b(c_R)\lambda_R\lambda_{R+1},$$

with $M_{\mathcal{I}} = \sum_{j \in \mathcal{I}} \lambda_j^2$ the window mass — only the two cut edges are unpaid. Relative to the compression, the completion changes exactly two kinds of term: the two cut-bond off-diagonals (already and only accounted for by the flux identity — not charged twice) and the two retained endpoint diagonals; nothing else changes. Each of the eight joint Bell terms has coefficient of modulus 1, and a joint term's diagonal is a product of two projector diagonals, so its change has modulus at most 2; the marginal coefficients contribute their absolute sum 4; hence the safe count $C_{\text{diag}} \leq 4 \cdot 1 + 8 \cdot 2 = 20$ and

$$0 \leq S - V_{\mathcal{I}} \leq (1 + 20R_{\max}/b_0) B_{\mathcal{I}}/(1 - T_{\mathcal{I}}),$$

with $T_{\mathcal{I}} := 1 - M_{\mathcal{I}}$ the omitted tail mass. Indeed, the ratio bounds and $b \geq b_0$ give

$$\lambda_{-L}^2 + \lambda_R^2 \leq (R_{\max}/b_0) B_{\mathcal{I}}.$$

The flux contributes $B_{\mathcal{I}}/M_{\mathcal{I}}$, while the two endpoint diagonals contribute at most $20(\lambda_{-L}^2 + \lambda_R^2)/M_{\mathcal{I}}$. This also distinguishes the coupling-weighted flux used here from the unweighted

boundary product used in [35]; their prefactors must not be interchanged. The fixed prefactor does not change the asymptotic exponent. Both $B_{\mathcal{I}}$ and $T_{\mathcal{I}}$ are bounded by $e^{-\kappa_- L + o(L)} + e^{-\kappa_+ R + o(R)}$, so balancing $L : R$ as $\kappa_+ : \kappa_-$ with $|\mathcal{I}| = d$ exactly:

$$S - S_d \leq S - V_{\mathcal{I}_d} \leq \exp(-\kappa_{\text{eff}} \cdot d + o(d)), \quad (18)$$

where the $o(d)$ absorbs the multiplicative constants, the integer rounding of L and R , and the ratio-to-tail conversion; therefore, for every $0 < \eta < \kappa_{\text{eff}}$, $D(\varepsilon) \leq (\kappa_{\text{eff}} - \eta)^{-1} \log(1/\varepsilon)$ eventually; letting $\eta \downarrow 0$ gives $\limsup D(\varepsilon)/\log(1/\varepsilon) \leq 1/\kappa_{\text{eff}} \leq 23.9010650$. This proves the upper half of Theorem 6; with the new lower proof in §5.1, Theorem 6 follows. $\square$

Remark 7 (Concrete dimensions). The geometric regime is visible at concrete sizes: the repository’s independently written verification script evaluates explicit strategies of the Pál–Vértesi family — a lower bound on S_d , not the truncation above — on the sampled ladder $d \in \{3, 4, 6, 8, 12, 16, 24, 33\}$, with values exceeding the two-qubit ceiling $1/4$ at $d = 24$ and $d = 33$ (high-precision evaluation with rigorous Rayleigh-quotient lower bounds reported by the script); by monotonicity of S_d , $S_d > 1/4$ for all $d \geq 24$. The script’s fixed parametrization is not Pál–Vértesi’s tuned family, so no conclusion about $d < 24$ follows from these evaluations, and nothing is claimed about the exact crossing point; in particular there is no tension with Pál–Vértesi’s own dimension-scan report, which concerns their tuned family.

5.3 Scope, and what is not claimed

Constants. The constants in the Θ are existential. The single explicit number 23.9010650 bounds the upper half’s constant only; it derives from the certified window and band algebra via an exact-rational chain, and is not claimed to be sharp. The lower half’s constants c, C are existential and no explicit values are claimed, so no explicit two-sided window on $D(\varepsilon)/\log(1/\varepsilon)$ is asserted; the $o(d)$ in (18) is not made explicit, and the Theorem’s inequalities are claimed for all sufficiently small ε .

Machine-checking boundary. Existing Lean theorems cover selected scalar and combinatorial facts in the earlier development. They do not formalize the new predictor, quantum response reduction, or existence of the signed potential. The elementary finite assembly alternative is formalized, but the complete analytic assembly is not. Exact graph checks are controls, not proofs of these analytic implications.

Inheritance. Both rate bounds use the critical storage and its compactly interior zero locus. Section 5.1 derives the endpoint positivity, collar, and approximate response estimates directly from that storage, reflection gluing, and $1/4 < S < 1/3$. The lower proof requires no strict graph structure and no spatial carrier. The upper proof still consumes the spatial-attainment certificate and its amended graph and orbit construction. This new lower proof neither revalidates those separate dependencies nor consumes the withdrawn paired-block step in the earlier rate certificate [34].

Priority, stated against the nearest prior art. Two qualifications are owed. First, the Ω -type dimension bounds already in the literature — for instance the $2^{\Omega(\varepsilon^{-1/8})}$ of [24] — attach to witnesses of the *distinct* separation $C_{qs} \neq C_{qa}$, and [24] gives a lower bound only, with no matching upper bound or tightness claim. Second, and more directly: Coladangelo and Stark remark, in the arXiv version of [4] immediately following their non-attainability theorem, that repeated application of their Schmidt-orbit maps exhibits an infinite geometric sequence and that “this can be used to obtain some quantitative bounds on the dimension required to

induce a correlation close to the ideal one”; they decline to prove it, on the stated ground that more useful bounds exist for the $C_{qs} \neq C_{qa}$ separation. That remark does not appear in the published version. We take it at face value: for their ideal state, whose Schmidt coefficients are exactly geometric, an $O(\log(1/\varepsilon))$ upper bound is a routine truncation argument, and their own discussion already names truncation to bounded-energy subspaces as the approximation mechanism. What we claim is therefore specific: a two-sided characterization for I_{3322} , with the upper half in §5.2 and the unrestricted lower half in §5.1. We note also what their method does not immediately give: the present carrier is *not* geometric but asymptotically geometric at two different rates at its two ends, which is why the constant is the harmonic-type combination $\kappa_{\text{eff}} = (1/\kappa_- + 1/\kappa_+)^{-1}$ rather than a single decay rate — that two-ended structure, forced by $r(t_+) < 1 < r(t_-)$, is the technical content of §5.2. These comparisons distinguish the scopes of the results; they are not a proof of an exhaustive priority claim.

6 Discussion: attainment and open carriers

The two theorems exhibit a sharp mechanism: *finiteness forces the equality transports to close (return), and closure caps the value strictly below S ; the value above the closed ceiling is carried only by an infinite open chain*. It is natural to ask how general this picture is. In every case known to us, a Bell functional with proven finite-dimensional attainment of its global supremum closes for an identifiable algebraic reason: bipartite correlator (XOR) functionals — including the chained inequalities — via Tsirelson’s Clifford construction [15, 16] (anticommutation is two-step closure); all $(2, 2; 2, 2)$ functionals, whose extreme points are realized by qubit strategies [14]; fixed CGLMP functionals, whose global optimum is attained at local dimension d (proven for $d = 3, 4$ [17]; we make no minimality claim); and, informally, those families whose sum-of-squares certificates force finite-dimensional algebraic relations on optimal strategies.

Conversely, both known explicit transport-class nonattainment objects are infinite open ladders: the Coladangelo–Stark state is an exactly geometric ladder whose nonattainment proof (Theorem 12 of the arXiv version of [4]) is an explicit no-return orbit argument on the Schmidt spectrum, and the Pál–Vértesi family is a half-infinite Jacobi chain with offset 2×2 blockings [3] — the same architecture, on smaller alphabets. Coladangelo and Stark suggested that the block-diagonal form of the conjectured optimal I_{3322} measurements, resembling that of their own construction, makes the inequality “potentially amenable to ideas and techniques from our work”; Theorem 1 may be read as an answer to that suggestion, reached by a transfer-operator route. Embezzlement-based nonattainment [22, 23, 24] is carried by the log-flat ladder $p_j \propto 1/j$ — an open chain of a different decay class; whether this heuristic correspondence with the operator-algebraic classification of embezzlement [25] can be made precise is an open question we do not pursue here.

Group-theoretic and type-based obstructions [12, 8, 10, 26] form a third family outside this transport picture: Slofstra’s undecidability of perfect-strategy existence for linear-system games [12], the RE-completeness of approximating the quantum value [26], and the arithmetical-hierarchy separation of [27], in which deciding whether the quantum value of a two-player non-local game equals 1 is Π_2 -complete (Mousavi–Nezhadi–Yuen) whereas deciding whether the commuting-operator value equals 1 is Π_1 (Slofstra, as restated there), precisely because the finite-dimensional supremum need not be attained. Together these imply that any general structural criterion must be restricted to a subclass.

For the specific I_{3322} equality module established here, a hypothetical finite-dimensional

maximizer would have a finite occupied response orbit; the resulting closure caps its value at $1/4 < S$. Extending this mechanism to a dichotomy for other two-outcome bipartite functionals is a conjectural direction, not a consequence of Theorem 1. Whether an appropriately defined finite recurrence conversely *suffices* for finite-dimensional attainment we do not know; nor do we have a definition of the critical transport for a general two-outcome functional, and supplying one is the first obstacle to testing it. We also note the marginal/correlator boundary: Tsirelson’s theorem covers correlators only, and it is exactly the marginal terms through which I_{3322} escapes it.

Finally, two cautions from the literature, both concerning the *hierarchy* sense of “attain” rather than the strategy sense used in this paper: non-saturation of finite levels of the Navascués–Pironio–Acín hierarchy [18] is not evidence of strategy-nonattainment — doubly-tilted CHSH functionals near the critical line $\alpha + \beta = 2$, where the boundary curvature degenerates, are attained at $d = 2$ yet are not saturated by NPA level 10, as reported in [28] with the level-10 figure due to Gigena and Kaniewski [32] — and the NPA hierarchy need not attain the commuting value at any finite level [29] (whose headline result is the undecidability of deciding $\omega_{qc} > 1/2$, in their notation for the commuting-operator value).

Theorem 6 is the quantitative face of this picture: the value carried by the open chain is approachable, but only at geometric cost, $D(\varepsilon) = \Theta(\log(1/\varepsilon))$. For contrast, the embezzlement-based witness of [24] — a witness of the distinct separation $C_{qs} \neq C_{qa}$ — requires local dimension $2^{\Omega(\varepsilon^{-1/8})}$; I_{3322} therefore sits at the opposite, logarithmic end of the known spectrum for the cost of approaching a value that is never attained.

The exact identification of S beyond the certified window remains open here. Pauwels’ Theorem 1 [5] proves that the full optimized finite Pál–Vértesi family has supremum S ; this is distinct from identifying the limit of a prescribed numerical parameter sequence. The amplitude-compatibility equation refuted in our correction history belonged to this project’s withdrawn construction. Its failure does not refute the Pál–Vértesi family. The spatial construction above uses an independently extracted probability carrier.

7 Methods and data

Verifying this paper’s claims

1. `pip install -r requirements.txt`, then `python tools/check_revision.py` — development checks for this revision, not a final release certificate. The frozen-release verifier and its scope are documented in `VERIFY.md`.
2. Theorem 1: `certificate/production/theorem-N-four-receipts-at-S/` — historical proof documents and exact-arithmetic guards. The current global-weld and occupied-component corrections are in Section 3 and `paper/REVISION-NOTES.md`.
3. Theorem 2: `certificate/production/theorem-S-spatial-attainment-at-S/` — proof documents and the amendment of 2026-08-07. The current parity-specific amplitude assignment is in Section 4.
4. Theorem 6, the rate $D(\varepsilon) = \Theta(\log(1/\varepsilon))$: the new lower proof in Section 5.1, and the constructive upper proof in Section 5.2, with its source record in `certificate/production/rate-theta-log/upper-U1G-bundle/`.
5. Revised lower bound: Section 5.1 contains the replacement proof; exact controls are in `certificate/production/lower-weighted-flow/`. The older lower bundle is preserved for history, not invoked for the quantitative conclusion.
6. Lean cores: `cd lean/I3322Kernel && lake build` — no `sorry`, standard axioms.
7. `VERIFY.md` at repository root: every claim mapped to its check and its analytic interfaces. `paper/CERTIFICATE-MAP.md` specifies correction precedence; `paper/BUILD.json` binds the current sources and PDF. The publication source manifest records the exact accompanying repository files.

Exact computation

The value-window certificates use rational arithmetic or outward-rounded interval arithmetic. Historical independent-engine checks and current replays have separate coverage: the current upper-window replay uses the inspected production arithmetic, while the lower quotient is reconstructed directly in rationals. The public verification guide records the scope of these checks; a saved receipt is not a new replay. Certificates and the correction record accompany the source repository [38]. The analytic steps are established in the proof documents; the guard scripts check algebraic identities and exact arithmetic only. A Lean 4 kernel [36] (`lean/I3322Kernel/`; with axiom reports in `AxiomCheck.lean`, using only standard axioms) machine-checks the quarter-ceiling chain of Section 3, the band algebra of §5.2, and the three combinatorial cores associated with the historical lower-bound route. The added `WeightedFlow` module checks selected finite accounting lemmas used in §5.1, including the all-edge balance and the residual estimate conditional on a bounded potential. Neither those lemmas nor the historical cores formalize the potential’s existence, the analytic estimates, or the complete quantum-to-flow argument.

Use of language models

Large language models were used throughout this work — for proof discovery, implementation, verification engineering, adversarial auditing, and editorial assistance — across multiple model families and separate, independently instantiated sessions, with the auditing sessions instructed to refute rather than confirm. The author conceived the research program and its hypotheses, set the verification standards and the preregistration and correction protocols, made all promotion, retraction, and publication decisions, and assumes sole responsibility for this work. The proof is an analytic, model-assisted argument for which the author remains responsible; exact computation does not by itself validate its analytic interfaces. The numerical certificates use rational or outward-rounded interval arithmetic, with the independently implemented checks documented in the repository. The Lean kernel checks the selected historical, upper-band, and finite weighted-flow lemmas specified above; the complete replacement lower proof remains analytic, not a Lean-certified theorem.

Data availability

Repository: <https://github.com/Apsiape/i3322-exact-wall>. The historical proofs and certificates are pinned to the archived release of record (v4.0.0), version DOI [10.5281/zenodo.22099128](https://doi.org/10.5281/zenodo.22099128); the concept DOI [10.5281/zenodo.21782008](https://doi.org/10.5281/zenodo.21782008) resolves to the latest release and is not the object cited here. The corrected proof is included in full in this paper and its accompanying `LOWER-BOUND-REVISION.tex`. The accompanying repository’s `paper/REVISION-NOTES.md` and `VERIFY.md` give correction precedence and reproduction instructions; `paper/BUILD.json` identifies the sources used for its PDF. The older archive does not contain this correction, and is cited for historical sources only. The historical certificate directories, all under `certificate/production/`, are

```
theorem-N-four-receipts-at-S/  
theorem-S-spatial-attainment-at-S/  
rate-theta-log/
```

Competing interests

None.

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